Demographic Transitions and Healthcare Demand in East Asia

DATASCI 350 - Data Science Computing | Final Project

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[Teammate 5]

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1 Introduction

East Asia is undergoing one of the most consequential demographic shifts in human history. Countries that were among the poorest in the world just fifty years ago have rapidly transformed into high-income economies — and in doing so, have accelerated through demographic transitions that took Western nations over a century to complete. The consequences for healthcare systems are profound and urgent.

This report analyses three World Bank World Development Indicators across eight East Asian and benchmark economies from 1990 to 2022: life expectancy at birth, under-5 mortality rate, and adolescent fertility rate. Together, these indicators capture the full arc of a country's demographic transition — from high mortality and high fertility societies toward low mortality, aging populations with shrinking youth cohorts.

Research Question: How have life expectancy, child mortality, and fertility trends diverged across East Asian economies from 1990 to 2022, and what do these demographic trajectories imply for future healthcare services demand?

1.1 Why This Matters for Healthcare

The demographic transition has direct implications for healthcare services demand. As under-5 mortality falls and life expectancy rises, healthcare systems shift their burden from acute paediatric and maternal care toward chronic disease management, elder care, and long-term care infrastructure. Countries like Japan, where over 29% of the population is already aged 65+, are experiencing this pressure today. South Korea, China, and Singapore are following closely behind.

For healthcare investors and policymakers, understanding where each country sits on this transition curve — and how fast it is moving — is essential for anticipating future demand patterns.

1.2 Summary of Main Findings

- **Japan leads all indicators** in 2022, with the highest life expectancy (84 years), lowest child mortality, and near-zero adolescent fertility
- **South Korea shows the most extreme fertility decline**, with the lowest adolescent fertility rate of all countries studied — a harbinger of its record-low national birth rate
- **China has achieved remarkable convergence** with high-income peers, gaining 10 years of life expectancy since 1990
- **The United States underperforms** relative to its income level, particularly in child mortality and the 2020-2021 COVID-19 life expectancy shock
- **Vietnam and Indonesia** are transitioning rapidly, but still face meaningful gaps in child survival compared to Northeast Asian peers

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2 Data Description

2.1 Dataset and Source

This analysis uses data from the World Bank's World Development Indicators (WDI) database, accessed via the `wbgapi` Python package. The WDI is the World Bank's primary collection of development indicators, compiled from officially recognised international sources.

Three indicators were selected:

Indicator	Code	Unit
Life expectancy at birth	`SP.DYN.LE00.IN`	Years
Under-5 mortality rate	`SH.DYN.MORT`	Per 1,000 live births
Adolescent fertility rate	`SP.ADO.TFRT`	Per 1,000 women aged 15-19

2.2 Countries and Classification

Eight countries were selected to represent the full spectrum of demographic development across East and Southeast Asia, with the United States included as a high-income benchmark:

Table 2: Countries included in the analysis

	Country	Code	Region	Demographic Group
0	Japan	JPN	East Asia	Aged Economy
1	South Korea	KOR	East Asia	Rapidly Aging
2	China	CHN	East Asia	Rapidly Aging
3	Singapore	SGP	Southeast Asia	Rapidly Aging
4	Thailand	THA	Southeast Asia	Transitioning
5	Vietnam	VNM	Southeast Asia	Transitioning
6	Indonesia	IDN	Southeast Asia	Young Economy
7	United States	USA	North America	Benchmark

2.3 Data Merging and Cleaning

Data for each indicator was downloaded separately from the World Bank API and merged into a unified long-format dataset. The merging process used `country_code` and `year` as join keys across all three indicator tables.

```
conn = sqlite3.connect('data/healthcare_demographics.db')

# Show the merge result
merge_check = pd.read_sql('''
    SELECT
        country_name,
        COUNT(*) as total_years,
```

```
        MIN(year) as first_year,
        MAX(year) as last_year,
        SUM(CASE WHEN life_expectancy IS NULL THEN 1 ELSE 0 END) as null_life_exp,
        SUM(CASE WHEN under5_mortality IS NULL THEN 1 ELSE 0 END) as null_mortality,
        SUM(CASE WHEN adolescent_fertility IS NULL THEN 1 ELSE 0 END) as null_fertility
    FROM clean_data
    GROUP BY country_name
    ORDER BY country_name
''', conn)
conn.close()
merge_check
```

Table 3: Data merging steps

	country_name	total_years	first_year	last_year	null_life_exp	null_mortality	null_fertility
0	China	33	1990	2022	0	0	0
1	Indonesia	33	1990	2022	0	0	0
2	Japan	33	1990	2022	0	0	0
3	Singapore	33	1990	2022	0	0	0
4	South Korea	33	1990	2022	0	0	0
5	Thailand	33	1990	2022	0	0	0
6	United States	33	1990	2022	0	0	0
7	Vietnam	33	1990	2022	0	0	0

The merged dataset contains **264 observations** (8 countries × 33 years) with **zero missing values** across all three indicators, confirming complete data coverage for the study period.

2.4 Summary Statistics

Table 4: Summary statistics by country (1990-2022)

	Country	Group	Avg Life Exp	Min Life Exp	Max Life Exp	Avg Mortality	Avg Fertili
0	Japan	Aged Economy	82.01	78.84	84.56	3.97	4.34
1	Singapore	Rapidly Aging	79.84	75.30	83.60	3.66	5.02
2	South Korea	Rapidly Aging	78.24	71.60	83.53	6.55	2.33
3	United States	Benchmark	77.29	75.21	78.84	8.03	37.96
4	China	Rapidly Aging	74.00	68.16	78.20	26.36	9.67
5	Thailand	Transitioning	73.30	68.92	77.61	18.82	45.44
6	Vietnam	Transitioning	72.79	69.03	75.38	28.66	36.54
7	Indonesia	Young Economy	67.35	63.25	70.92	43.42	48.46

3 Data Analysis

3.1 Life Expectancy Trends (1990–2022)

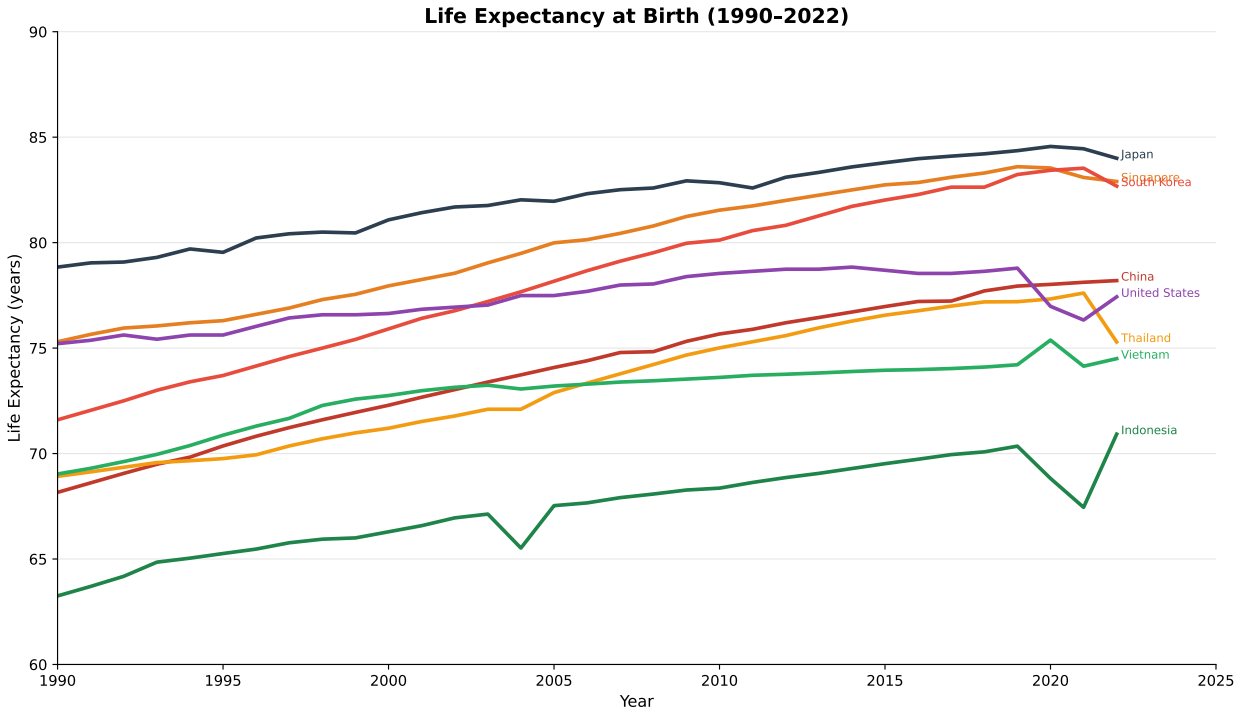


Figure 1: Life expectancy at birth by country (1990-2022)

Life expectancy rose across all eight countries between 1990 and 2022, reflecting improvements in healthcare access, nutrition, and living standards. Japan maintained the highest life expectancy throughout, reaching 84 years by 2022. The most notable convergence is China's trajectory — rising from 68 years in 1990 to 78 years in 2022, closing the gap with high-income peers by approximately 10 years in just three decades.

The United States exhibits a notable decline in 2020–2021, reflecting the disproportionate impact of COVID-19 on American life expectancy compared to East Asian peers. Indonesia shows a dip around 2004, consistent with the mortality impact of the Indian Ocean tsunami.

3.2 Under-5 Mortality Trends (1990–2022)

Under-5 mortality declined dramatically across all countries, with the steepest improvements in Vietnam and Indonesia — both countries more than halved their child mortality rates between 1990 and 2022. Japan and Singapore already had near-zero child mortality by 1990 and maintained those levels throughout the period, leaving little room for further improvement.

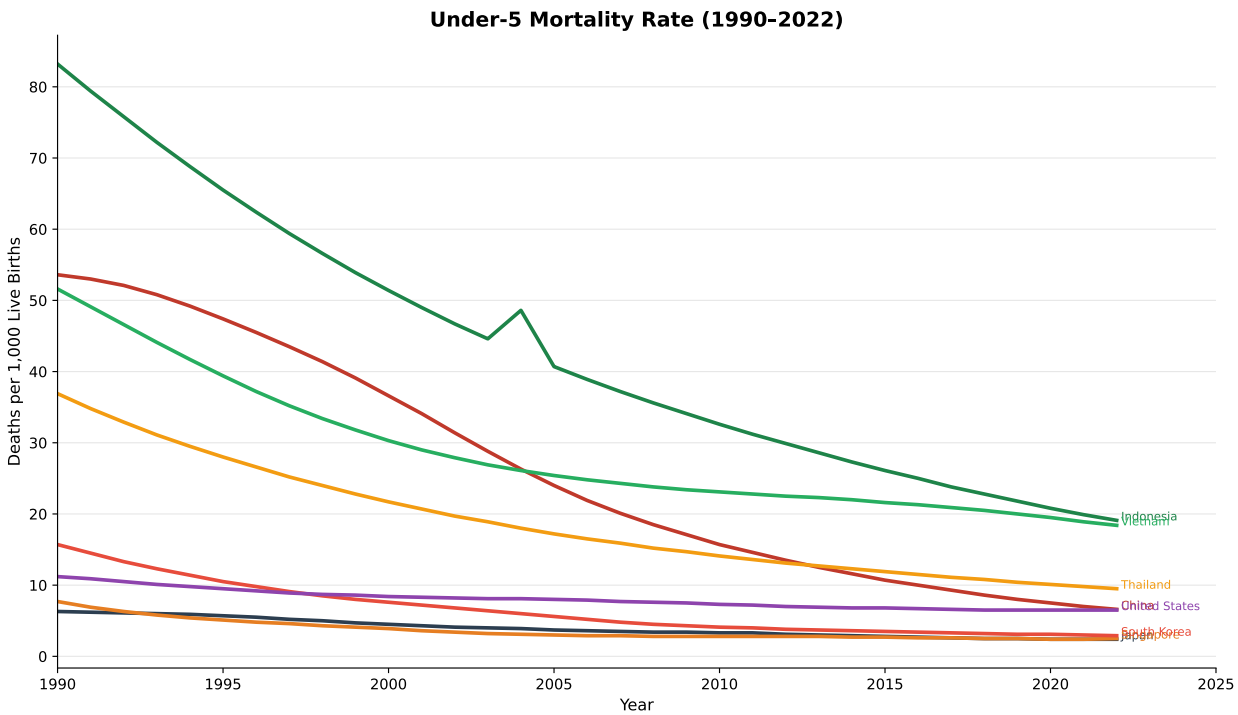


Figure 2: Under-5 mortality rate by country (1990-2022)

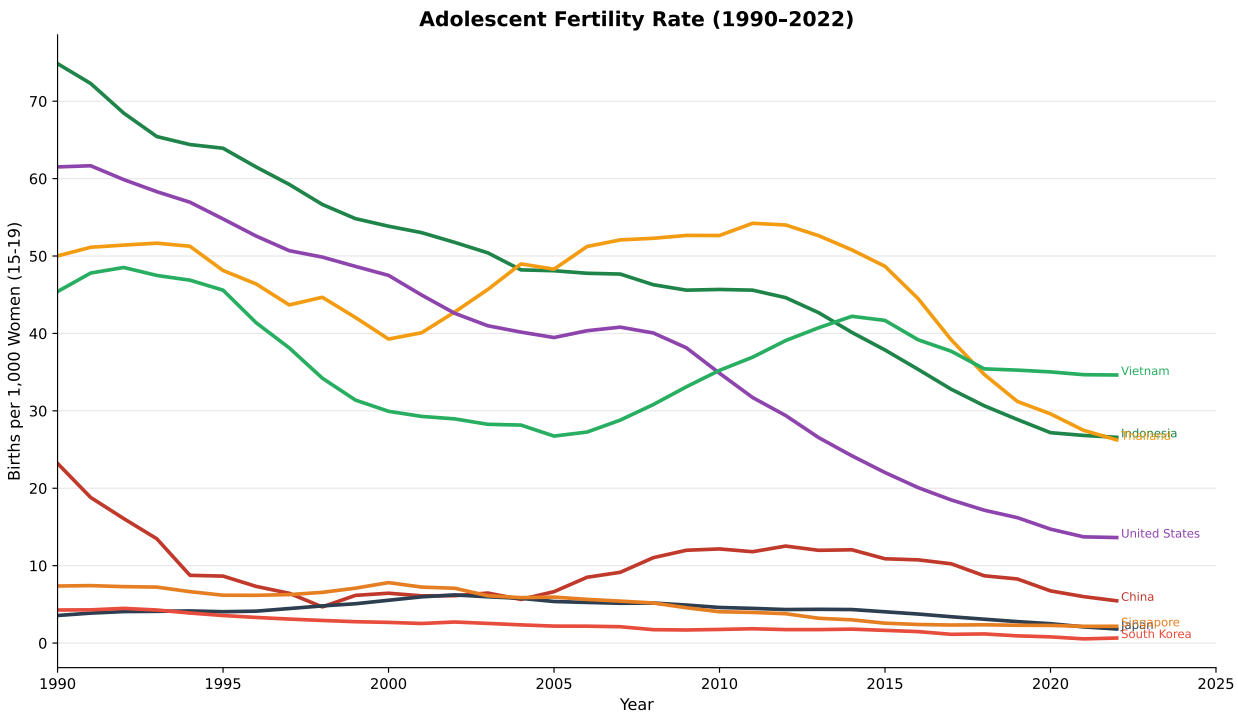


Figure 3: Adolescent fertility rate by country (1990-2022)

3.3 Adolescent Fertility Trends (1990–2022)

Adolescent fertility declined across all countries, with Indonesia showing the most dramatic reduction — falling from approximately 60 births per 1,000 teenage women in 1990 to under 25 by 2022. South Korea and Japan maintained near-zero adolescent fertility throughout the period, consistent with their status as the world's most demographically aged economies.

3.4 Demographic Health Scorecard (2022)

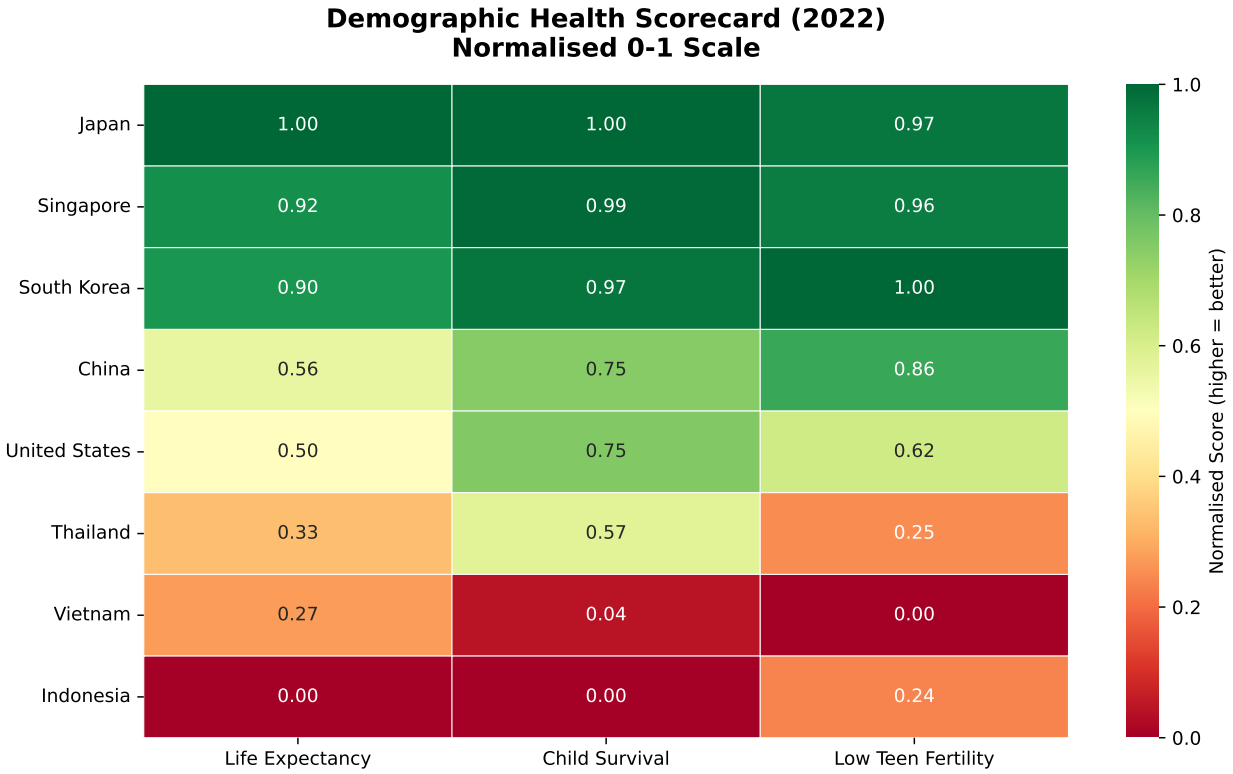


Figure 4: Demographic health scorecard — normalised scores by country (2022)

The scorecard normalises all three indicators to a 0-1 scale where higher values represent better outcomes. Japan, Singapore, and South Korea dominate the top tier across all three dimensions. The United States scores surprisingly low relative to its income level — particularly on life expectancy (0.50) — reflecting structural inefficiencies in the American healthcare system compared to East Asian peers.

3.5 Country Improvement (1990–2022)

South Korea leads all countries in life expectancy gain, adding approximately 12 years between 1990 and 2022, reflecting its rapid economic development and healthcare investment over the period. China and Indonesia follow closely, each gaining around 8–10 years. For under-5 mortality reduction, China and Indonesia show the largest absolute improvements, reducing child deaths by

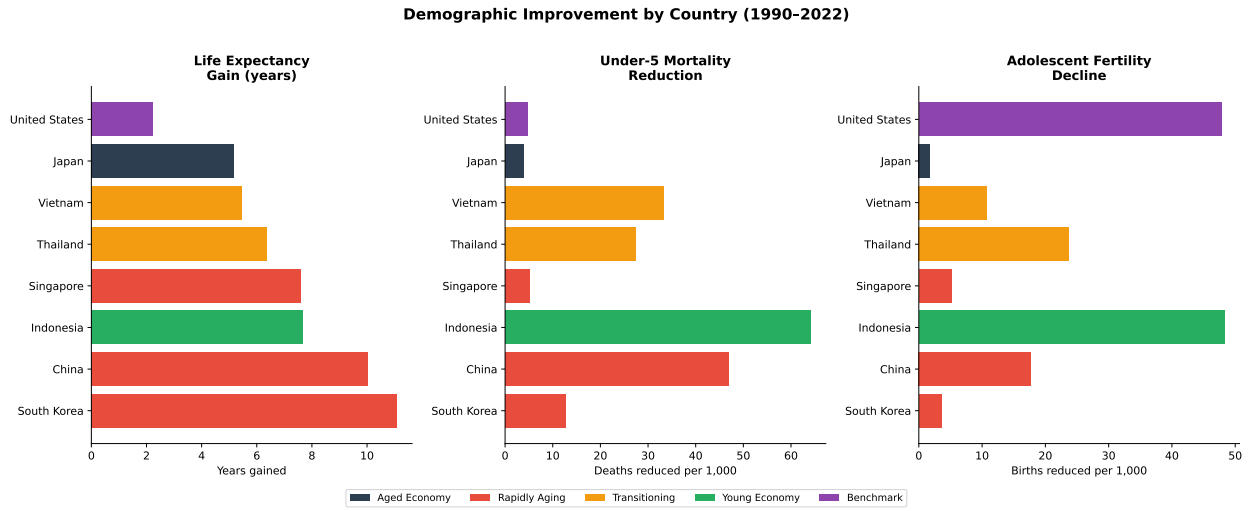


Figure 5: Absolute improvement in each indicator from 1990 to 2022

approximately 50–60 per 1,000 live births — unsurprising given both countries started from much higher baselines in 1990, leaving Japan and Singapore little room to improve. Adolescent fertility decline was most dramatic in Indonesia and the United States, both reducing by around 40–50 births per 1,000 teenage women, while South Korea and Japan showed minimal decline as their rates were already near zero in 1990.

3.6 Life Expectancy vs Child Mortality (2022)

The scatter plot confirms a strong negative relationship between child mortality and life expectancy — countries that have eliminated child deaths also have the longest-lived populations. Japan sits at the frontier, with essentially zero preventable child mortality and the world's highest life expectancy. The United States is a clear outlier, with higher child mortality than China despite significantly higher per-capita healthcare spending.

3.7 Additional Relationship: Fertility and Life Expectancy

There is also a clear negative relationship between adolescent fertility and life expectancy. Countries with lower adolescent fertility rates tend to have higher life expectancy, reflecting more advanced demographic transitions.

This pattern reinforces the broader finding that improvements in public health, education, and economic development are closely linked to declining fertility and increasing longevity. As countries progress through the demographic transition, healthcare demand shifts away from maternal and child health services toward aging-related care, chronic disease management, and long-term care infrastructure.

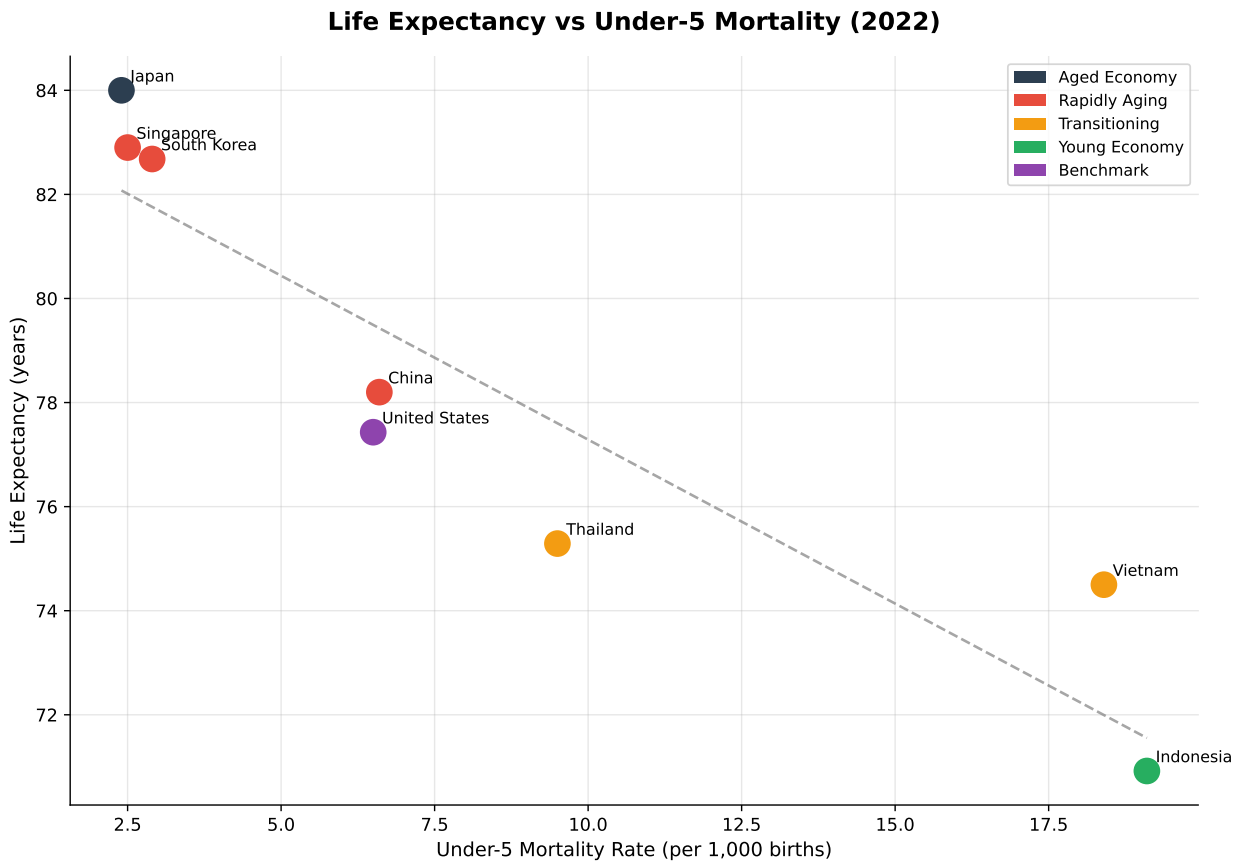


Figure 6: Relationship between life expectancy and under-5 mortality (2022)

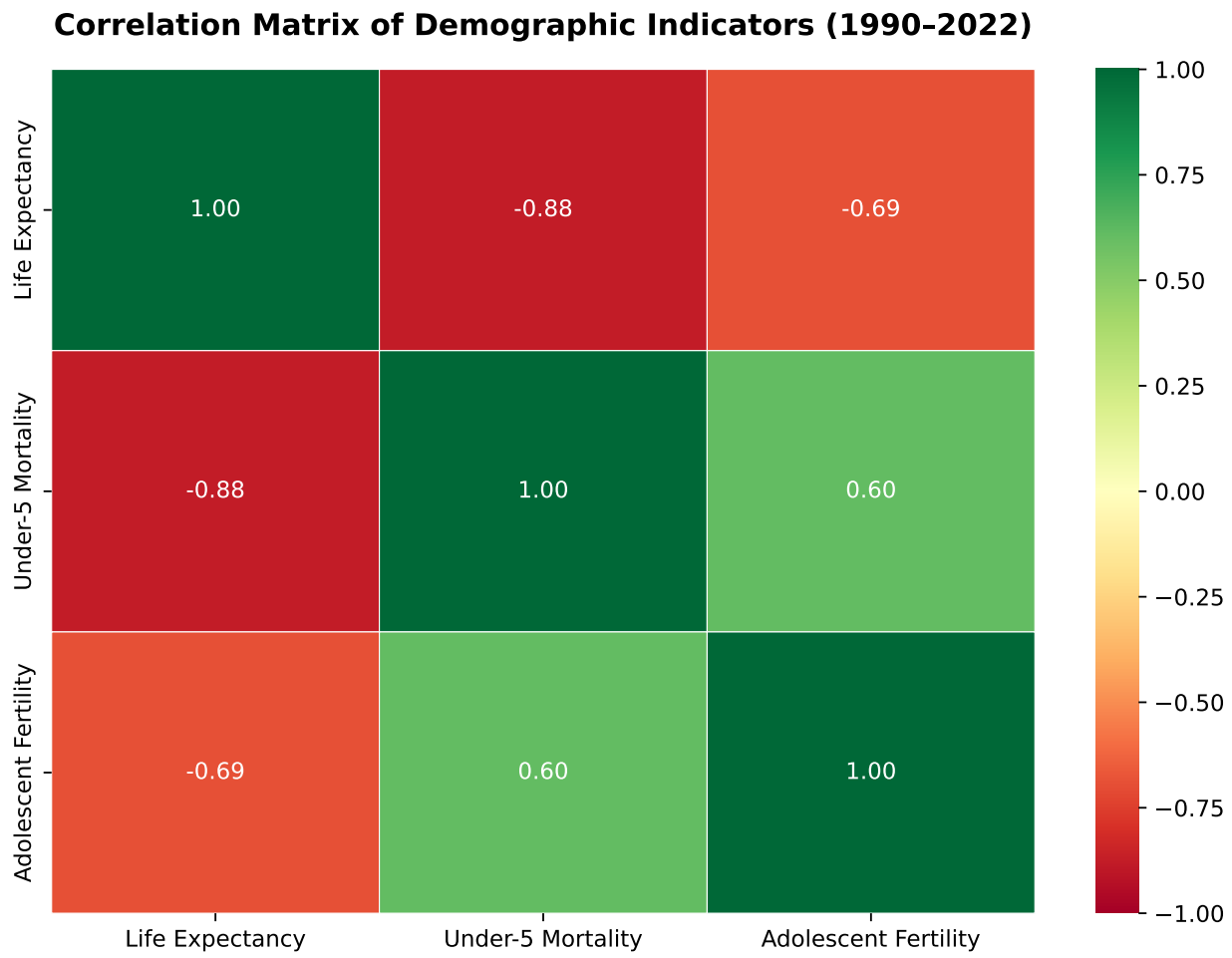
3.8 Correlation Matrix of Demographic Indicators (1990-2022)

```
corr_matrix = df[['life_expectancy', 'under5_mortality', 'adolescent_fertility']].corr()

corr_matrix.index = ['Life Expectancy', 'Under-5 Mortality', 'Adolescent Fertility']
corr_matrix.columns = ['Life Expectancy', 'Under-5 Mortality', 'Adolescent Fertility']

fig, ax = plt.subplots(figsize=(8, 6))
sns.heatmap(corr_matrix,
            annot=True,
            fmt='.2f',
            cmap='RdYlGn',
            vmin=-1, vmax=1,
            ax=ax,
            linewidths=0.5)

ax.set_title('Correlation Matrix of Demographic Indicators (1990-2022)',
             fontsize=13, fontweight='bold', pad=15)
plt.tight_layout()
plt.show()
```



The correlation matrix confirms the strong relationships between all three indicators. Life expectancy shows a strong negative correlation with both under-5 mortality and adolescent fertility, reflecting that countries with higher child mortality and teen birth rates tend to have shorter-lived populations. Under-5 mortality and adolescent fertility are also positively correlated with each other, consistent with both being markers of earlier-stage demographic transitions.

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4 Results and Discussion

4.1 Key Finding 1: Northeast Asia Has Completed Its Demographic Transition

Japan, South Korea, and Singapore have effectively completed their demographic transitions. Child mortality is near zero, life expectancy exceeds 82 years, and adolescent fertility has collapsed. The healthcare demand implications are clear: these countries now face acute pressure on elder care capacity, chronic disease management, and long-term care infrastructure. Japan's healthcare system — already the most elder-focused in the world — represents a preview of what South Korea and Singapore will face within the next decade.

4.2 Key Finding 2: China's Rapid Convergence Creates Immense Demand

China's demographic trajectory is arguably the most consequential finding in this analysis. Adding 10 years of life expectancy in 32 years — while simultaneously suppressing fertility through the one-child policy — has created a demographic structure that will generate unprecedented demand for healthcare services. With 1.4 billion people aging simultaneously and a healthcare system that is still building capacity, China represents the single largest healthcare services opportunity in the world over the next two decades.

4.3 Key Finding 3: Southeast Asia Is at an Inflection Point

Vietnam, Thailand, and Indonesia are undergoing rapid demographic transitions but have not yet reached the healthcare system pressures of Northeast Asia. This creates a window of opportunity: these countries are simultaneously still addressing maternal and child health needs while beginning to invest in elder care infrastructure. Healthcare systems and investors that move early in these markets — particularly in Vietnam and Thailand where transitions are most advanced — are positioned to capture significant long-term value.

4.4 Key Finding 4: The United States as a Cautionary Benchmark

The United States consistently underperforms relative to its income level across all three indicators. Its child mortality rate is higher than China's in 2022, its life expectancy dropped dramatically during COVID-19 while East Asian peers maintained stability, and its adolescent fertility rate lags significantly behind East Asian peers. This suggests structural differences in healthcare system design and public health capacity that go beyond simple resource availability.

4.5 Limitations

- The analysis uses three indicators as proxies for broader demographic and healthcare system dynamics. A more complete analysis would incorporate age structure data, healthcare expenditure, and disease burden indicators.
- Country-level aggregates mask significant within-country variation, particularly for large countries like China and Indonesia where rural-urban disparities are substantial.
- The study period ends in 2022. South Korea's birth rate has continued to fall dramatically in 2023-2024, and the implications of this are not captured here.

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5 Conclusion

This analysis documents a clear and measurable demographic divergence across East Asia from 1990 to 2022. Northeast Asian economies — Japan, South Korea, and Singapore — have completed their demographic transitions and now face immediate, structural demand for elder care and chronic disease services. China is converging rapidly toward this group while managing a population of

unprecedented scale. Southeast Asian economies remain in transition, creating both ongoing needs for maternal and child healthcare and emerging demand for elder care infrastructure.

For healthcare services investors, these findings point toward a multi-decade demand wave that is already underway in Northeast Asia and approaching rapidly in Southeast Asia. The demographic data analysed here — life expectancy, child survival, and fertility trends — are not merely academic indicators. They are the leading signals of where healthcare infrastructure investment will be needed, at what scale, and on what timeline.

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6 References